

Exam #1- CHEM 330- Fall 2025

Name..... ANSWERS

A) Multiple Choice Section (30 min). 24 questions. Only one correct answer. 2.5 points each.

1) Aspirin (acetylsalicylic acid) is a weak acid with a pK_a of 3.4. Stomach acid has a pH of approximately 1.4. What is the ratio of the protonated form of aspirin to the deprotonated form of aspirin at equilibrium in the stomach? [Rule: if $\log_{10}(x)=b$, then $x=10^b$]

- A) 1:10
- B) 10:1
- C) 1:1
- D) 1:100
- ☒ E) 100:1

$$\begin{aligned} \text{ASP-H} &\rightleftharpoons \text{ASP}^- + \text{H}^+ & pK_a &= 3.4 & pH_{\text{stomach}} &= 1.4 \\ pH &= pK_a + \log \left(\frac{[A^-]}{[AH]} \right) \\ 1.4 &= 3.4 + \log \left(\frac{[A^-]}{[AH]} \right) \\ 1.4 - 3.4 &= \log \left(\frac{[A^-]}{[AH]} \right) = -2 \\ \text{so, } \frac{[A^-]}{[AH]} &= 10^{-2} & \frac{[AH]}{[A^-]} &= \frac{1}{10^{-2}} = 10^2 \end{aligned}$$

2) Alzheimer's disease is caused by the accumulation of the amyloid β ($A\beta$) protein in brain regions serving memory and cognition. Interestingly, $A\beta_{42}$, a peptide that contains just two extra amino acids at the C-terminus than $A\beta_{40}$, is considered the "toxic" species in Alzheimer's disease because it tends to aggregate more than its shorter counterpart, $A\beta_{40}$. The two additional C-terminal residues in $A\beta_{42}$ are isoleucine and alanine. Based on your understanding of amino acid properties, how might these two extra residues lead to enhanced aggregation of the peptide?

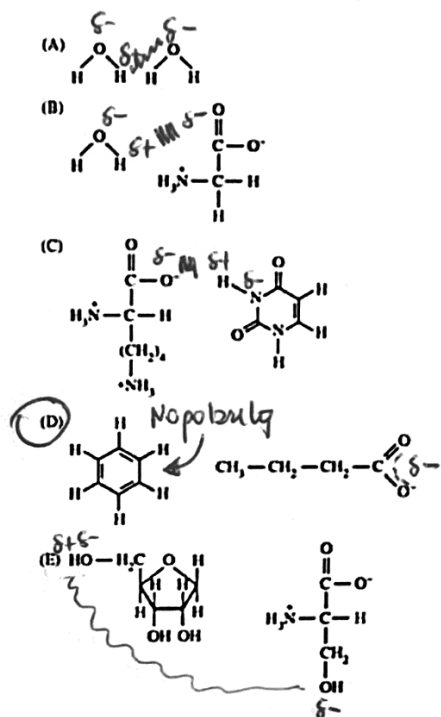
- A) The additional two residues may participate in hydrogen bonding that facilitate protein aggregation *No hydrogen bonds between ile and ala*
- ☒ B) the additional two residues may participate in hydrophobic interactions that facilitate protein aggregation *He, both are non-polar*
- C) the additional two residues may participate in electrostatic interactions that facilitate protein aggregation *No polarity, C-C and C-H bonds*
- D) the additional two residues may participate in disulfide bonding to facilitate protein aggregation *NO S.*
- E) none of the above

3) In a mixture of the five globular proteins listed below, which should elute first in size exclusion (gel- filtration) chromatography using Sephadex G-100? The fractionation range of Sephadex G-100 is 4,000-150,000 Da.

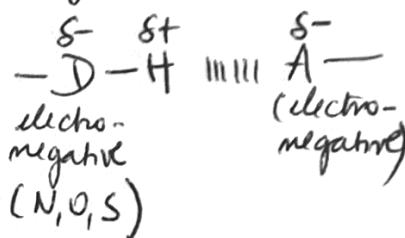
- A) cytochrome c MW = 13,000 Da
- ☒ B) RNA polymerase MW = 450,000 Da
- C) immunoglobulin G MW = 145,000 Da
- D) ribonuclease A MW = 13,700 Da
- E) serum albumin MW = 68,500 Da

Bigger molecules elute first-
Because the biggest is (B) at 450 kDa, it will elute first in the volume outside the pore (V_o).
Because (C) has a size lower than the exclusion limit, it will elute second (partially included in the gel pore)-

4) Which of the following pairs of molecules could NOT hydrogen-bond with each other?

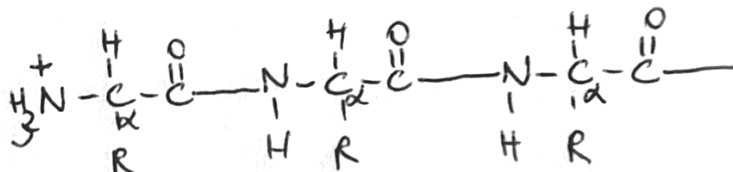


you just need this structure:

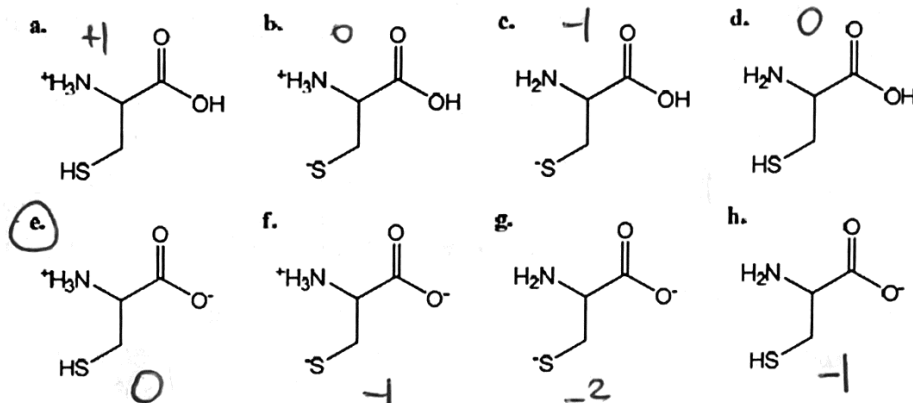


5) Which of the following best represents the backbone arrangement of two peptide bonds?

- A) $C_\alpha-N-C_\alpha-C-C_\alpha-N-C_\alpha-C$
 B) $C_\alpha-N-C-C-N-C_\alpha$
 C) $C-N-C_\alpha-C_\alpha-C-N$
 (D) $C_\alpha-C-N-C_\alpha-C-N$
 E) $C_\alpha-C_\alpha-C-N-C_\alpha-C_\alpha-C$



6) Given the ionization states of the amino acid cysteine, which one is the zwitterion?



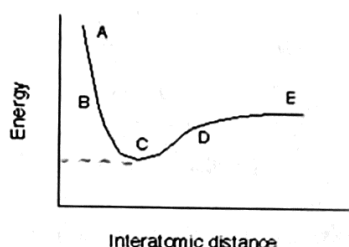
charge zero-

So, b, d, e-

pKs are 1.7 (α COOH),
 10.8 (α NH₃⁺), 8.3 (R)-
 So, α COOH should
 deprotonate first,
 R chain α cond, α NH₃⁺
 last- option b de-
 protonated R chain
 first (impossible);

option b deprotonated α NH₃⁺ first (impossible).
 option e deprotonated α COOH first (OK)

7) The maximal attractive force of a van der Waals interaction occurs at which point on the following energy diagram?



Maximal attractive force should release the most energy into the surroundings, stabilizing and decreasing the energy of the system - so, C (lowest energy point of the system)

- A) A
- B) B
- ☒ C) C
- D) D
- E) E

8) Which of the following statements correctly describes the structural organization of α -helices and β -sheets in proteins?

A) In α -helices, hydrogen bonds form between backbone groups of different polypeptide strands, while in β -sheets they form within one strand. (the opposite)

B) In α -helices, side chains alternate above and below the helical axis, while in β -sheets they project radially outward from the axis. (the opposite)

☒ C) In α -helices, hydrogen bonds form between the backbone C=O of residue i and the N-H of residue $i+4$, while in β -sheets hydrogen bonds form between backbone groups of neighboring strands. Yes!

D) Both α -helices and β -sheets have side chains projecting into the interior, stabilizing the secondary structure through hydrophobic packing. Neither project into the interior

E) Both α -helices and β -sheets repeat every 3.6 residues, forming a helical twist stabilized by R-group hydrogen bonding. only α helices - part of the sheet is the repeating structure in β -sheets

9) A sequence of amino acids in a certain protein is found to be -Ser-Gly-Pro-Gly-. The sequence is most probably part of a(n):

A) antiparallel β sheet.

B) parallel β sheet.

C) α -helix.

D) α -sheet.

☒ E) β -turn. common in β -turns.

too flexible, poor side-chain stabilization - Disfavors regular secondary structures -
 forces a turn; can't hydrogen-bond (N lost its H in the peptide bond) - Destabilizes secondary structures that depend on them for stabilization -

10) The weakest non-covalent interactions are:

A) ionic interactions

B) hydrogen bonds

C) hydrophobic interactions

☒ D) van der Waals interactions

E) disulfide bonds

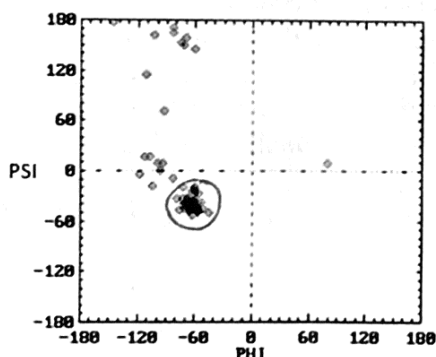
van der Waals < hydrophobic < hydrophobic < ionic < covalent

11) Which of the following amino acids is the most soluble in water at pH 7.0?

- (A) Glutamate $-\text{COO}^-$ *it depends on the polarity of the R side chain at pH 7.*
 B) Tryptophan $-\text{NH}$
 C) Leucine $-\text{CH}_3$
 D) Tyrosine $-\text{OH}$
 E) Phenylalanine $-\text{OH}$

12) The Ramachandran plot for a particular protein is shown below. This protein is composed primarily of:

- (A) α -helices
 B) parallel β -sheets
 C) antiparallel β -sheets
 D) a collagen-like triple helix, made of a repetitive primary sequence of glycine-proline-hydroxyproline



$\sim (-60, -60)$ - Partial folding, in the same direction (α helix)

13) What would you expect about the formation of an α -helix for a segment of a protein chain that contains lysine approximately every fourth residue with all other residues being mostly hydrophobic?

- A) Helix formation would never occur regardless of pH.
 (B) Helix formation would be favored at high pH.
 C) Helix formation would be favored at low pH.
 D) Helix formation would be favored at neutral pH.

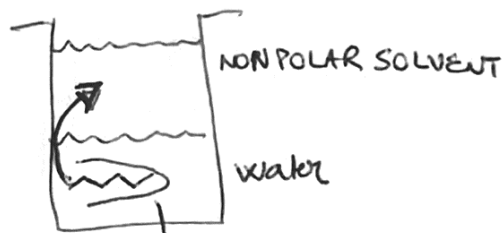
lys --- lys --- lys --- lys
 $\text{NH}_3^+ \dots \text{NH}_3^+ \dots \text{NH}_3^+ \dots \text{NH}_3^+$
 repulsion (close in space in α helix) - Need to deprotonate them at high pH to avoid this
 $-\text{NH}_3^+ + \text{OH}^- \rightarrow -\text{NH}_2 + \text{H}_2\text{O}$

14) In the energetics of the process of transferring hydrocarbons from water to a nonpolar solvent, the factor $T\Delta S_{\text{sys}}$ is commonly:

- A) Negative
 (B) Positive
 C) Unmeasurable
 D) Unimportant
 E) Assumed to be zero

absolute temperature, always positive

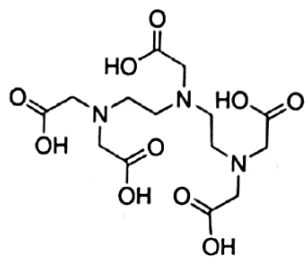
$$\Delta G = \Delta H - T\Delta S$$



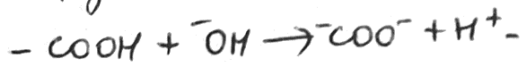
$S_{\text{of the system}}$ increases as water pockets are removed, and replaced by "flickering clusters" - The system is defined as the water-

"water pockets", low entropy

15) Dr. Caro needs to prepare a concentrated aqueous solution of the iron chelator DTPA (diethylenetriaminepentaacetic acid). He purchases item D6518 from Sigma-Aldrich:



If it doesn't dissolve in water, probably low polarity, weak acid, only partially ionized - How can I increase their ionization? Titrate them with strong base:



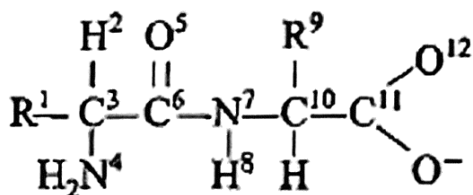
Once ionized, it should be more soluble by ion-dipole interactions with water. A weak base is not the best option, you need a high $[OH^-]$ to titrate all the $-COOH$ -

DTPA is a pentaprotic acid with pKa values of 1.8, 2.6, 4.4, 8.8, and 10.4. When Dr. Caro tries to dissolve solid DTPA in distilled water, he is unsuccessful.

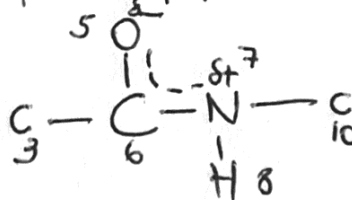
What would you recommend he do to dissolve it?

- A) Add a strong acid such as HCl until DTPA dissolves
- ☒ B) Add a strong base such as NaOH until DTPA dissolves
- C) Add a weak acid such as acetic acid until DTPA dissolves
- D) Add a weak base such as NH_3 until DTPA dissolves
- E) Do not use DTPA; there is no way to dissolve it in water

16) If these were in a protein in a trans configuration, which atoms would lie in a plane?



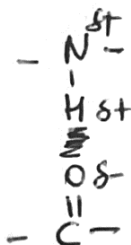
Any atom directly connected to the planar peptide bond:



- A) 2,3,4,5,6,7
- B) 3,4,5,6,7,10
- C) 3,4,6,7,8,10
- ☒ D) 3,5,6,7,8,10
- E) 3,6,7,8,9,10

17) The major bonds or interactions that stabilize the secondary structures of proteins, between atoms of the polypeptide backbone are:

- A) covalent bonds
- ☒ B) hydrogen bonds
- C) hydrophobic interactions
- D) ionic bonds
- E) van der Waals interactions



A) Histidine $\text{N}^{\oplus}\text{H}$
 B) Glutamate $-\text{COO}^-$
 C) Serine $-\text{OH}_{\delta+}$
 D) Phenylalanine $-\text{C}_6\text{H}_5$
 E) Aspartate $-\text{COO}^-$

NON-POLAR, hydrophobic clustering- At pH 7,

- (A) Acetic acid / acetate: $pK_a = 4.8$ 3.8-5.8
 (B) Phosphate ($H_2PO_4^-/HPO_4^{2-}$): $pK_a = 7.2$ 6.2-8.2
 (C) Tris (tris(hydroxymethyl)aminomethane): $pK_a = 8.1$ 7.1-9.1
 (D) Bicarbonate (H_2CO_3/HCO_3^-): $pK_a = 6.3$ 5.3-7.3
- Buffering range: $pK_a \pm 1$

contrast with L-dopa, it causes deficiency of white blood cells and thus susceptibility to infections. Which of the structures shown below is levodopa?

Handwritten notes above the structures:

- Structure A: rotate in plane: D
- Structure B: rotate in plane: D
- Structure C: rotate in plane: D
- Structure D: rotate in plane: D

Handwritten note at the top right: L-amino-acid (Fischer)

Handwritten note at the top center: $\text{H}_3\text{N}^+ - \text{C} - \text{H}$ (Fischer projection)

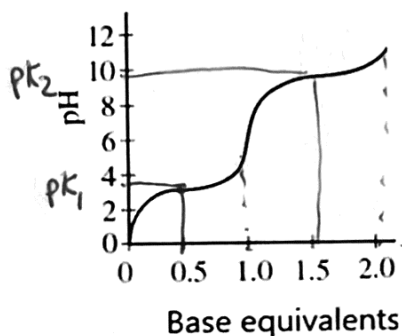
Chemical structures shown:

- A: [O-]C(=O)[C@H](N)Cc1ccc(O)c(O)c1
- B: [O-]C(=O)[C@H](N)Cc1ccc(O)c(O)c1
- C: [O-]C(=O)[C@H](N)Cc1ccc(O)c(O)c1
- D: [O-]C(=O)[C@H](N)Cc1ccc(O)c(O)c1

A. Val
 B. Asp
 C. Ala
 D. Ile
 E. Phe

LEU ——— VAL: NON POLAR ALIPHATIC
 NON POLAR ALIPHATIC
 ASP: POLAR, NEG. CHARGED
 ALA: NON POLAR ALIPHATIC
 ILE: NON POLAR ALIPHATIC
 PHE: NON POLAR AROMATIC

22) The most likely amino acid that would yield the above titration curve below is:

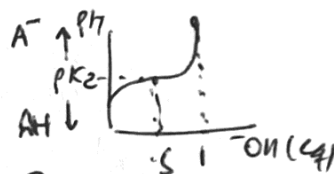
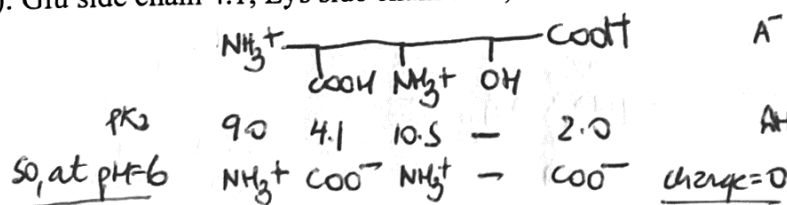


only two pK_2 s at 0-12 range
one acid, one basic -
probably αCOOH , αNH_3^+ respectively
NO $pK_3 = pK_R$ -

- A) Arginine $pK_R \approx 12$
- B) Glutamic acid $pK_R \approx 4$
- ☒ C) Glycine $pK_R \approx 6$
- D) Histidine $pK_R \approx 6$
- E) Tyrosine $pK_R \approx 10$

23) Consider the tripeptide Glu-Lys-Ser at pH 6.0. What is the net charge?
pKa (approx.): Glu side chain 4.1; Lys side chain 10.5; N-terminus 9.0; C-terminus 2.0.

- A) -2
- B) -1
- ☒ C) 0
- D) +1

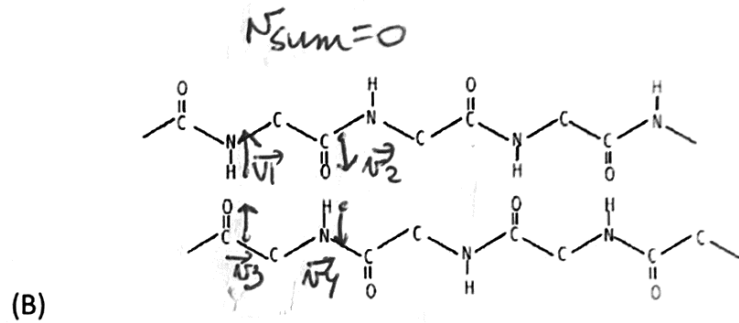
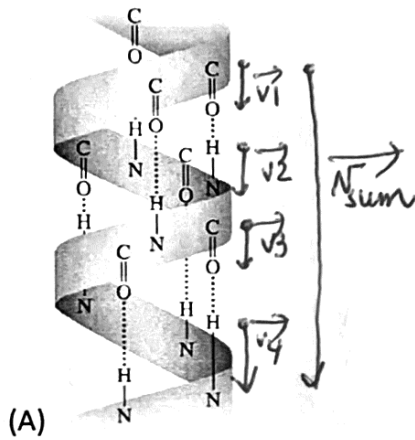


24) Which of the following best describes fibrous proteins such as collagen and keratin?

- A) They are highly flexible, soluble in water, and have irregular primary sequences
- ☒ B) They are rigid, insoluble in water, and have repetitive primary sequences
- C) They are compact, soluble in water, and often act as enzymes
- D) They are globular, flexible, and stabilized mainly by hydrophobic cores

C) Short Answers Section (20 min). 4 questions. 10 points each

1) Look at the structure of a typical alpha helix (A), and a typical antiparallel beta sheet (B):



Alpha helices have a molecular dipole moment, but beta sheets do not. Explain why.

Molecular dipole moments are the vector sum of bond dipole moments. The polar bonds are C=O and N-H bonds. $\delta^+ \delta^-$ and $\delta^- \delta^+$

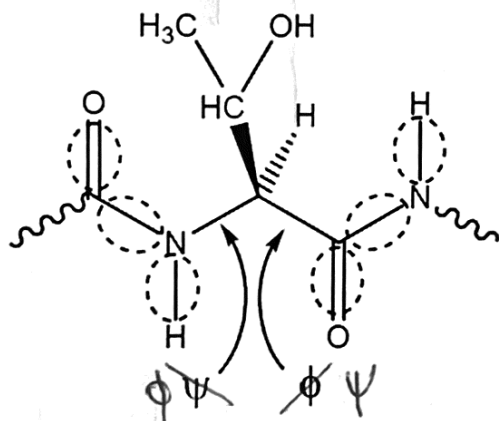
The vectors point from + to -, and are parallel to the bond.

In an alpha helix, all the bond dipole moments point in the same direction, from the N terminus to the C terminus. So, there is a vector sum, and a molecular dipole moment.

In beta sheets, the bond dipole moments alternate directions, so their vector sum is zero, and there is no molecular dipole moment.

Draw a segment of a polypeptide chain with:

- One student gives the following answer:



→ Not tyrosine, but threonine

→ This is a totally extended conformation, no (ϕ, ψ) are $(180^\circ; 180^\circ)$, not $(180^\circ; 0)$

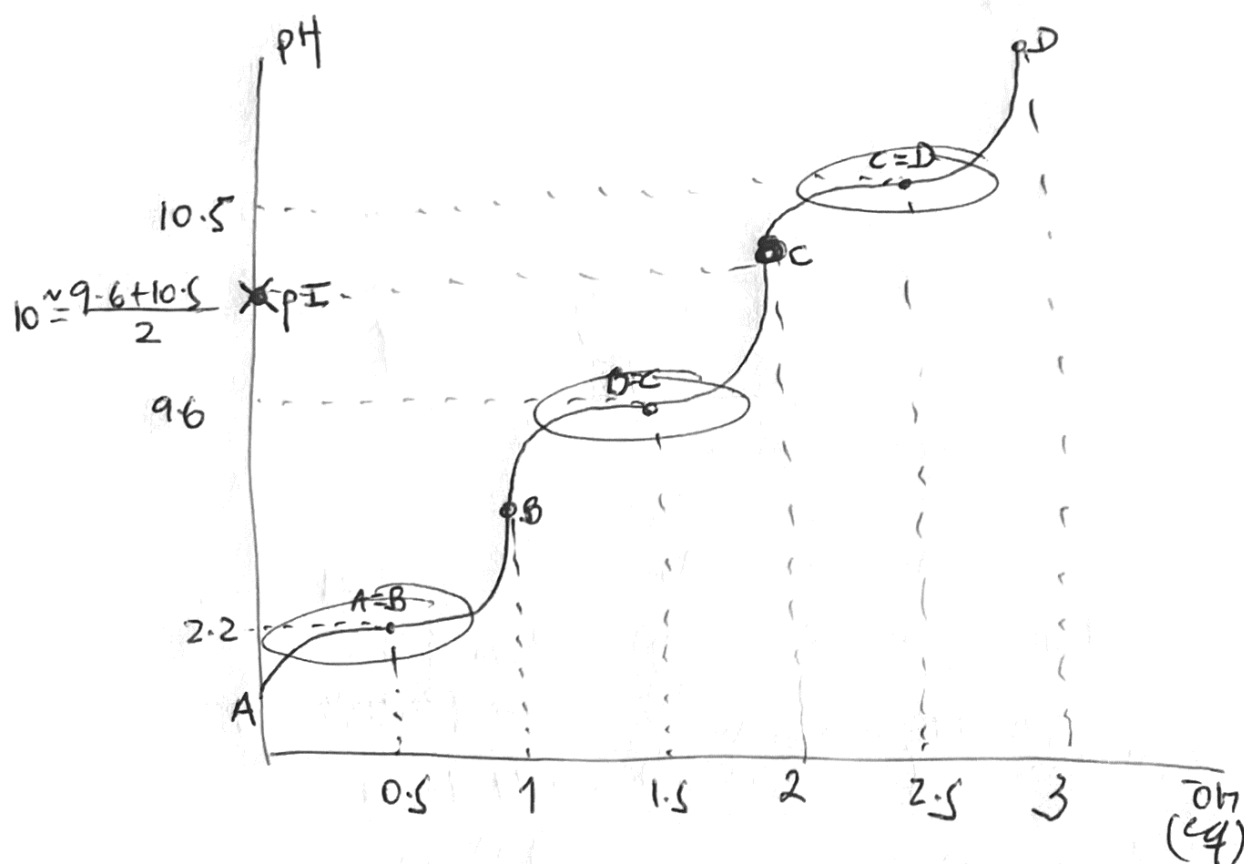
→ The N-H bond (2 of them) can rotate, and they were circled - The C and C-N are OK, they can't rotate (partially)

→ The anions are incorrect (see correction above) double bonds

3) Sketch the titration curve for the tripeptide Val-Lys-Ser.

- Label the x-axis as "OH⁻ equivalents added". Clearly mark the first, second, and third equivalence points as 1, 2, and 3.
- Indicate the corresponding half-equivalence points (0.5, 1.5, 2.5).
- Label the y-axis as "pH".
- Mark on the y-axis the approximate pK_a values at their half-equivalence points
- Circle on the curve the buffering regions.
- Mark on the y-axis the isoelectric point (pI) of the tripeptide.

NOT IN SCALE!

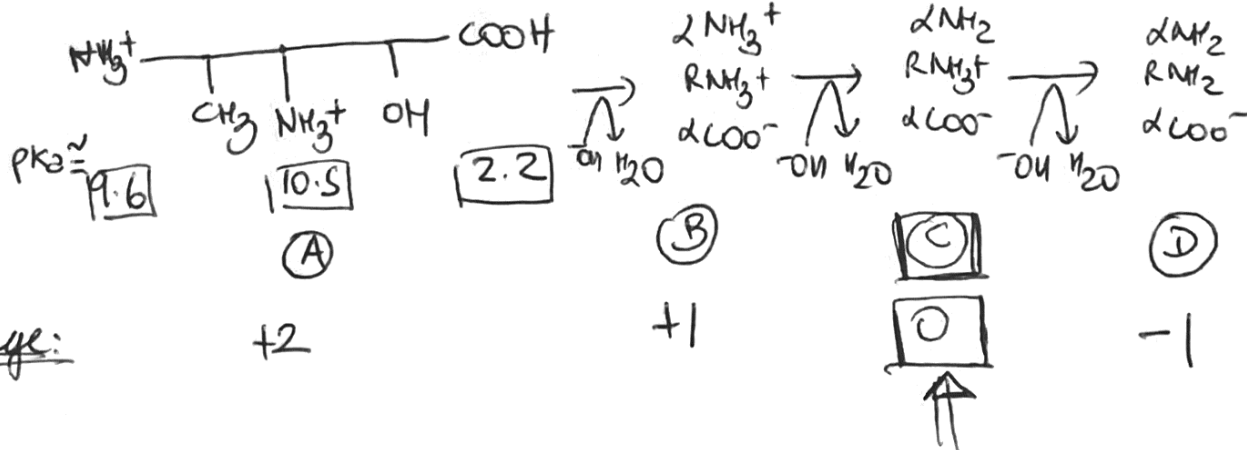


C is the zwitterion:

$pK_2 = 2.2$

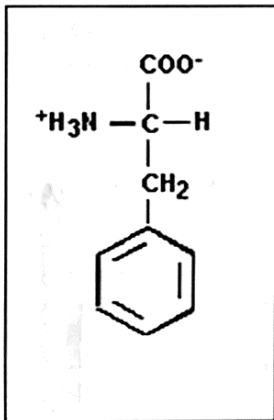
$pK_2 = 9.6$

$pK_2 = 10.5$



4) Fill the boxes below:

Structure at pH 7.0

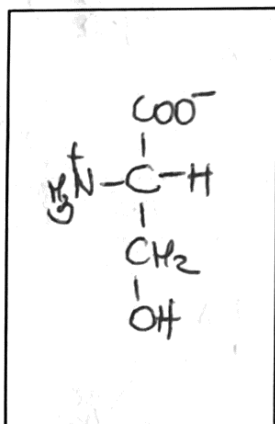


Three letter code

PHE

Classification

AROMATIC
NON POLAR



SER

POLAR
UNCHARGED

Amino acid	α -COOH pK_a	α -NH ₃ ⁺ pK_a	R-group pK_a
Gly	2.3	9.6	-
Ala	2.4	9.7	-
Val	2.3	9.6	-
Leu	2.4	9.6	-
Iso	2.4	9.7	-
Met	2.4	9.2	-
Pro	2.1	10.6	-
Phe	1.8	9.1	-
Trp	2.4	9.4	-
Ser	2.2	9.2	-
Thr	2.6	10.4	-
Tyr	2.2	9.1	10.1
Cys	1.7	10.8	8.3
Asn	2.0	8.8	-
Gln	2.2	9.1	-
Asp	2.1	9.8	3.9
Glu	2.2	8.8	4.3
Lys	2.2	9.0	10.5
Arg	2.2	9.0	12.5
His	2.4	9.2	6.0